Rules:

- *půl X-é* = ‘half past $(X - 1)$’
- *čtvrt na X* = ‘quarter past $(X - 1)$’
- *za X minut Y* = ‘X minutes to Y’, where Y can be a full hour (o’clock) or any of the two structures above.

Problem 9.7a 1. ‘23 past 5’ = ‘7 minutes to [half past five]’ = ‘7 minutes to [half towards 6]’ = *za sedm minut půl šesté*

2. ‘10 past 9’ = ‘5 minutes to [quarter past nine]’ = ‘5 minutes to [quarter towards 10]’ = *za pět minut čtvrt na deset*

In this case, overcounting is used for all the structures, but, as mentioned above, there are languages in which only certain structures use overcounting. For example, in German, overcounting is only used in the half-past constructions – *halb X* = ‘half past $(X - 1)$’.

Problem 9.8

Swahili (Nilai Sarda, PLO 2014)

Here are some Swahili phrases and their English translations *in random order*:

1.	*jumamosi, saa moja usiku*	A.	‘Sunday, 1.00 AM’
2.	*jumapili, saa tatu na robu asubuhi*	B.	‘Sunday, 7.30 AM’
3.	*jumamosi, saa saba usiku*	C.	‘Sunday, 9.15 AM’
4.	*jumamosi, saa mbili na robu usiku*	D.	‘Tuesday, 12.15 PM’
5.	*jumanne, saa tano na nusu usiku*	E.	‘Tuesday, 11.30 PM’
6.	*jumanne, saa sita na robu asubuhi*	F.	‘Saturday, 10.30 AM’
7.	*jumamosi, saa nne na nusu asubuhi*	G.	‘Saturday, 7.00 PM’
8.	*jumapili, saa moja na nusu asubuhi*	H.	‘Saturday, 8.15 PM’

Problem 9.8a Determine the correct correspondences.

Problem 9.8b Translate into English:

9. *jumatano, saa moja na robu asubuhi*
10. *jumapili, saa nne na nusu asubuhi*

Problem 9.8c Translate into Swahili:

11. 'Monday, 12.15 AM' 12. 'Monday, 10.00 PM'

Solution

The first step is noticing the structure of the phrases. All of them start with a word separated by a comma, which has the prefix *juma-*. This most likely represents the day (it is unlikely that all that variety of times would use the same one-word structure). Therefore, the structure after the comma must be the time. For this, we notice two types of structures:

saa X usiku and *saa X na {robu/nusu} {usiku/asubuhi}*

We can assume that the simplest structure refers to the o'clock times, which is also reinforced by the fact that we only have two such structures in both Swahili and English. Thus:

jumamosi, saa moja usiku	'Sunday, 1.00 AM'
jumamosi, saa saba usiku	'Saturday, 7.00 PM'

Curiously, in Swahili, the same name seems to be used to express different days in English. We will have to explain why that is in due course.

Among the remaining examples, only one more contains the hour 7 (although it is AM, instead of PM), and only one other example starts with either *saa saba* or *saa moja* (since we do not know the correspondence between the two). Since phrase 8 contains *saa moja*, we deduce that it corresponds to translation B. Moreover, we deduce that *saa moja* means 7 o'clock, so we can make three correspondences:

1.	*jumamosi, saa moja usiku*	G.	'Saturday, 7.00 PM'
3.	*jumamosi, saa saba usiku*	A.	'Sunday, 1.00 AM'
8.	*jumapili, saa moja na nusu asubuhi*	B.	'Sunday, 7.30 AM'

Based on example 8, we can infer that *na nusu* means 'half past'. Moreover, most likely, *usiku* and *asubuhi* correspond, in some way, to the notion of AM and PM, but it is not a one-to-one correspondence, since *usiku* is used for both 7 PM and 1 AM.

We only have two more examples which contain *na nusu* (5 and 7), and the English times correspond to 10.30 AM and 11.30 PM. Since *usiku* is used for both 7 PM and 1 AM, we can assume that it will also be used for 11.30 PM, so that *usiku* suggests the idea of 'evening/night'.

We get two more correspondences: